COSAI

COALITION FOR SECURE AI

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Abstract

Summary of the technical purpose of the document.

[By creating structured opportunities for interaction and feedback, it aims to foster cooperation and produce outcomes that are practical, measurable, and valued by all involved.]

A Practical Guide to MCP Security

Why MCP Security Matters Now

Actionable Security Controls

Introduction

Since its emergence a year ago, MCP has rapidly established itself as the protocol for transmitting structured context between AI agents and services. Given the growing importance and attack surface of MCP and agentic systems, it is imperative that deployment specific security threats are identified and improvements are made to address the challenges and ambiguities inherent in MCP implementations....

Section 1 Title

MCP is an open standard developed by Anthropic and a growing open source community that provides a structured framework for connecting large language models and AI agents to external tools, data sources, and services. MCP addresses a fundamental challenge in agentic AI systems: how to enable models to access and interact with real-world resources dynamically while maintaining security and reliability through standardization. MCP simplifies AI application integration with databases, APIs, file systems, web services, and other external resources reducing the need for custom integrations for each tool or service.

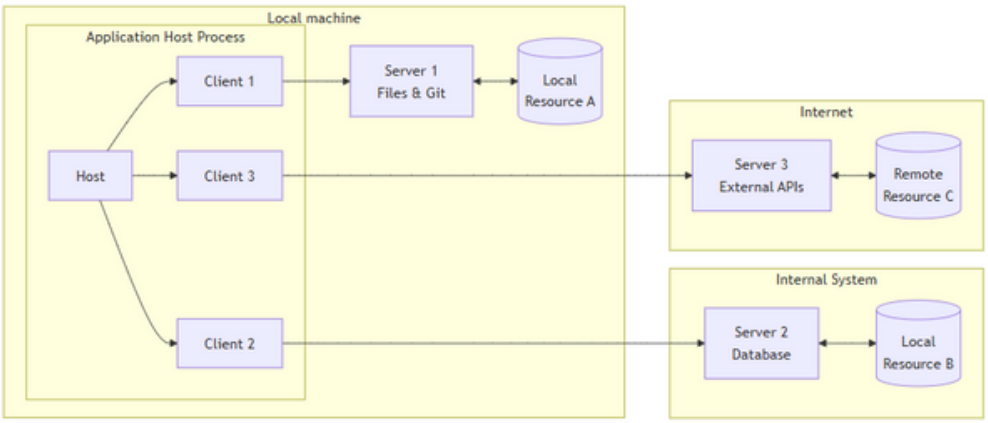
MCP operates through a client-server architecture where AI applications (hosts) use MCP clients to establish connections with MCP servers that expose specific capabilities such as tools, resources, and prompts. The protocol defines standardized methods for:

- discovering available capabilities including tools or services
- invoking parametrized services
- accessing data resources

MCP supports multiple transport mechanisms including standard I/O (stdio) for local processes and Streamable HTTP for networked communications, enabling flexible deployment scenarios from local development environments to distributed cloud architectures.[2]

Section 1.1 Title

MCP follows a client-server architecture where host applications (such as AI assistants, IDEs, or workflow automation tools) use MCP clients to connect to local or remote MCP servers. Each client-server connection operates as a dedicated, stateful session that begins with initialization and capability negotiation. Communication is built on JSON-RPC, which defines the message format and protocol semantics including lifecycle management and core primitives (tools, resources, and prompts). The transport layer handles the delivery of these JSON-RPC messages between clients and servers, supporting stdio for local processes and Streamable HTTP for remote servers.



Section 1.2 Title

Takeaways and Conclusion

MCP adoption is accelerating, and security must keep pace. Our analysis reveals common vulnerabilities in deployments that lack adequate authentication, session management, and supply chain controls. Incidents in adjacent AI systems demonstrate these are active threats, not theoretical concerns.

Organizations deploying MCP-based systems must develop defense-in-depth strategies including zero-trust architectures, hardware-based isolation through trusted execution environments, rigorous supply chain vetting, and continuous monitoring. Securing MCP deployments requires coordinated effort across developers, organizations, and protocol maintainers — investment in security architecture now will pay dividends as agentic systems become more prevalent.

References

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Appendix

CoSAI Focus

CoSAI is an OASIS Open Project, bringing together an open ecosystem of AI and security experts from industry-leading organizations. The project is dedicated to sharing best practices for secure AI deployment and collaborating on AI security research and product development. The scope of CoSAI is specifically focused on the secure building, integration, deployment, and operation of AI systems, with an emphasis on mitigating security risks unique to AI technologies. Other aspects of Trustworthy AI are deemed important but beyond the scope of the project including, ethics, fairness, explainability, bias detection, safety, consumer privacy, misinformation, hallucinations, deep fakes, or content safety concerns like hateful or abusive content, malware, or phishing generation. By concentrating on developing robust measures, best practices, and guidelines to safeguard AI systems against unauthorized access, tampering, or misuse, CoSAI aims to contribute to the responsible development and deployment of resilient, secure AI technologies.

Guidelines on usage of more advanced AI systems (e.g. large language models (LLMs), multi-modal language models. etc) for drafting documents for OASIS CoSAI:

tl;dr: CoSAI contributions are actions performed by humans, who are responsible for the content of those contributions, based on their signed OASIS iCLA (and eCLA, if applicable). [Each contributor must confirm whether they are entitled to donate that material under the applicable open source license; OASIS and the CoSAI Project do not separately confirm that.] Each contributor is responsible for ensuring that all contributions comply with these AI use guidelines, including disclosure of any use of AI in contributions.

- Selection of AI systems: CoSAI recommends the use of reputable AI systems (lowering the risk of inadvertently incorporating infringing material).
- Model constraints: Currently, CoSAI or OASIS are not required to have a contract or financial agreement for using AI systems from specific vendors. However, CoSAI editors should consider employing varying tools to avoid potential fairness concerns among vendors.
- IP infringement: It is the responsibility of the individual who subscribes/prompts and receives a response from an AI system to confirm they have the right to repost and donate the content to OASIS under our rules.
- Transparency: CoSAI's goal will be to maintain transparency throughout the process by documenting substantial use of AI systems whenever possible (e.g., the prompts and the AI system used), and to ensure that all content, regardless of production by human or AI systems, was reviewed and edited by human experts. This helps build trust in the standards development process and ensures accountability.

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